

Chapter 15.8: The Change of Variables Formula

Mechanics

1. Find the Jacobian determinant of the transformation $x = e^{-r} \sin(\theta)$, $y = e^r \cos(\theta)$.
2. An ellipse in the plane has the equation $\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1$ for positive numbers a and b . Use a change of variables to compute the area of this ellipse. *[Note: Without a change of variables, you would likely need a trig substitution to do this!]*
3. Compute $\iint_R e^{x+y} dA$, where R is the region given by the inequality $|x| + |y| \leq 1$. *[Hint: Sketching the region should reveal a natural choice of transformation to use.]*
4. Use the change-of-variables formula to justify the cylindrical and spherical volume elements $dx dy dz = r dr d\theta dz = \rho^2 \sin \varphi d\rho d\theta d\varphi$ we learned last week.
5. Solve the system

$$u = 2x - 3y, v = -x + y$$

for x and y in terms of u and v . Then find the value of the Jacobian and find the image of the parallelogram R in the xy -plane with boundaries $x = -3$, $x = 0$, $y = x$, and $y = x + 1$ under this transformation. Sketch the transformed region in the uv -plane. Use your results to rewrite the integral

$$\iint_R 2(x - y) dx dy$$

as an integral in uv -coordinates.

Extensions

6. Find equations for a transformation $\mathbf{T}$ that maps a rectangular region S in the uv -plane whose sides are parallel to the u - and v -axes onto the region R bounded by the hyperbolas $y = 1/x$, $y = 3/x$ and the lines $y = x$, $y = 3x$ in the first quadrant.
7. Use a change of variables to compute

$$\iint_R xy dA,$$

where R is the region in the first quadrant bounded by the lines $y = x$ and $y = 3x$ and the hyperbolas $xy = 1$, $xy = 3$. *[Hint: Consider the above problem.]*